

Gretsch Chet Atkins Circuit Analysis



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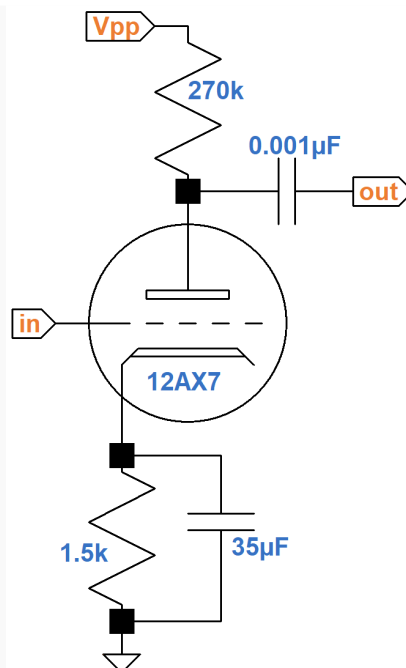
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The Gretsch Chet Atkins amplifier has a unique architecture: an active 12AX7 mixer driven by independent channels, each in turn having separate bass and treble channels that drive a passive mixer. This article focuses on these aspects of the design. It also derives a formula for the gain of the active mixer.

Preamp

Unlike that vast majority of guitar amplifiers, the Gretsch Chet Atkins uses the two triodes of a 12AX7 to drive bass and treble controls independently. Each triode forms a traditional common-cathode voltage amplifier. The treble channel preamp is as shown here.



The coupling capacitor driving the bass control is large enough to act as a short circuit for all audio frequencies. For the treble driver, it is only $0.001\mu\text{F}$, which creates significant bass attenuation. The cathode resistor for the treble driver is $1.5\text{k}\Omega$ and is bypassed by a $35\mu\text{F}$ capacitor. This effectively shorts the cathode to ground for all audio frequencies and creates 38dB gain. The cathode resistor in the bass driver is larger, $2.2\text{k}\Omega$, and unbypassed. According to the [Preamp Gain and Output Impedance calculator](#), this reduces the unloaded gain to 34dB.

unloaded voltage gain	48.67 (33.75dB)
output impedance ($\text{k}\Omega$)	138.58

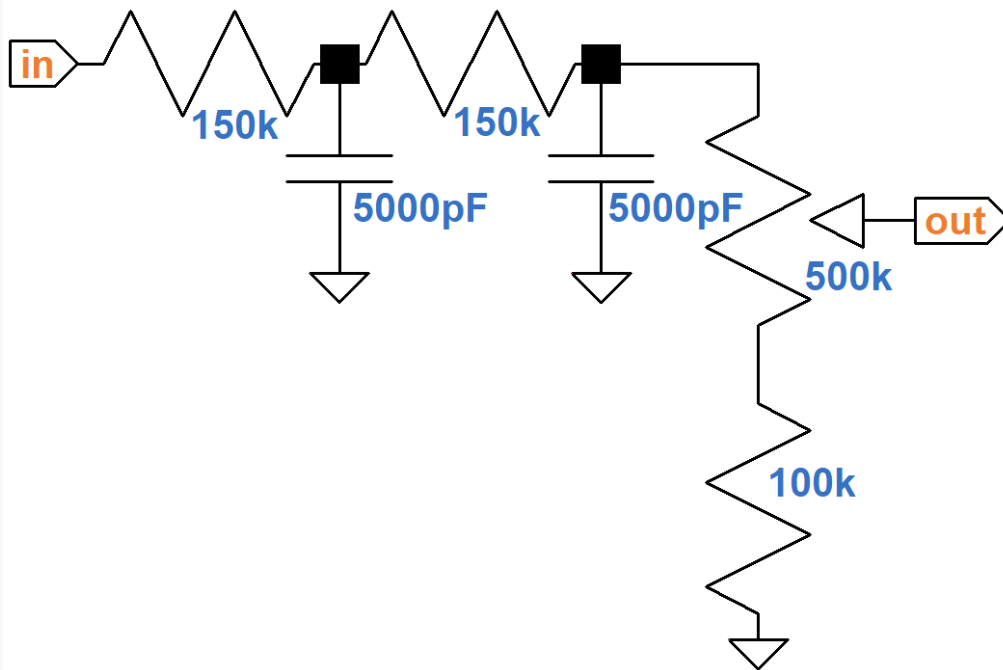
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tube type	12AX7 ▼
plate resistor R_L ($\text{k}\Omega$)	270
cathode resistor R_K ($\text{k}\Omega$)	2.2
Is R_K bypassed by a capacitor C_K?	☐ yes ☒ no

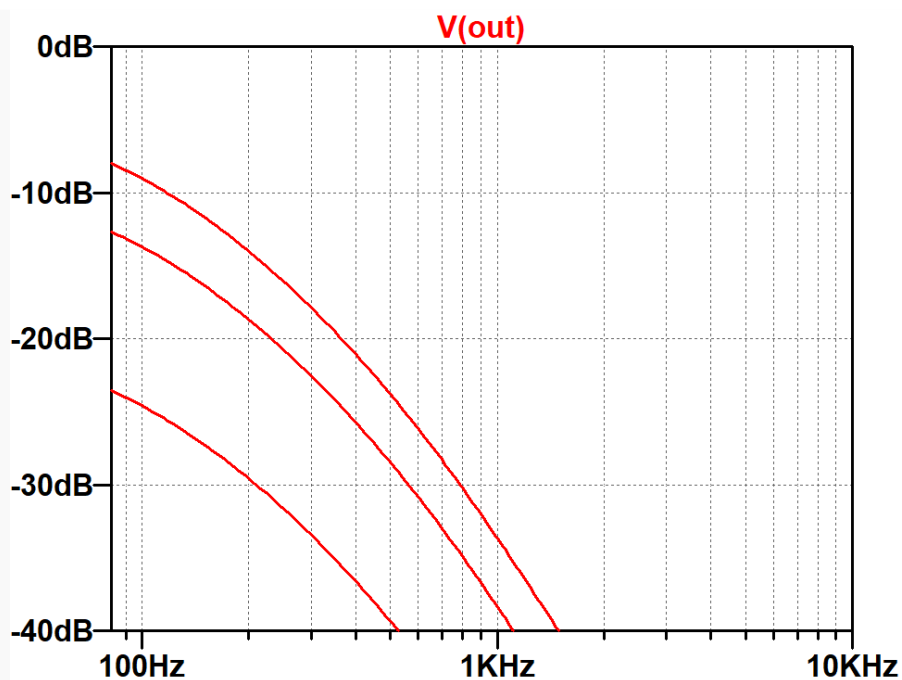
The treble driver's output impedance¹ is about $51\text{k}\Omega$. The calculator shows that the bass driver has an output impedance of $139\text{k}\Omega$. In many circumstances a high output impedance is a disadvantage, because the stage is more likely to be loaded down by the current it is forced to supply to drive the next stage. In this case, however, the output impedance is a desirable artifact. The design, in fact, places a $150\text{k}\Omega$ resistor in series with the output impedance driving the bass channel, as shown below.

Bass Control

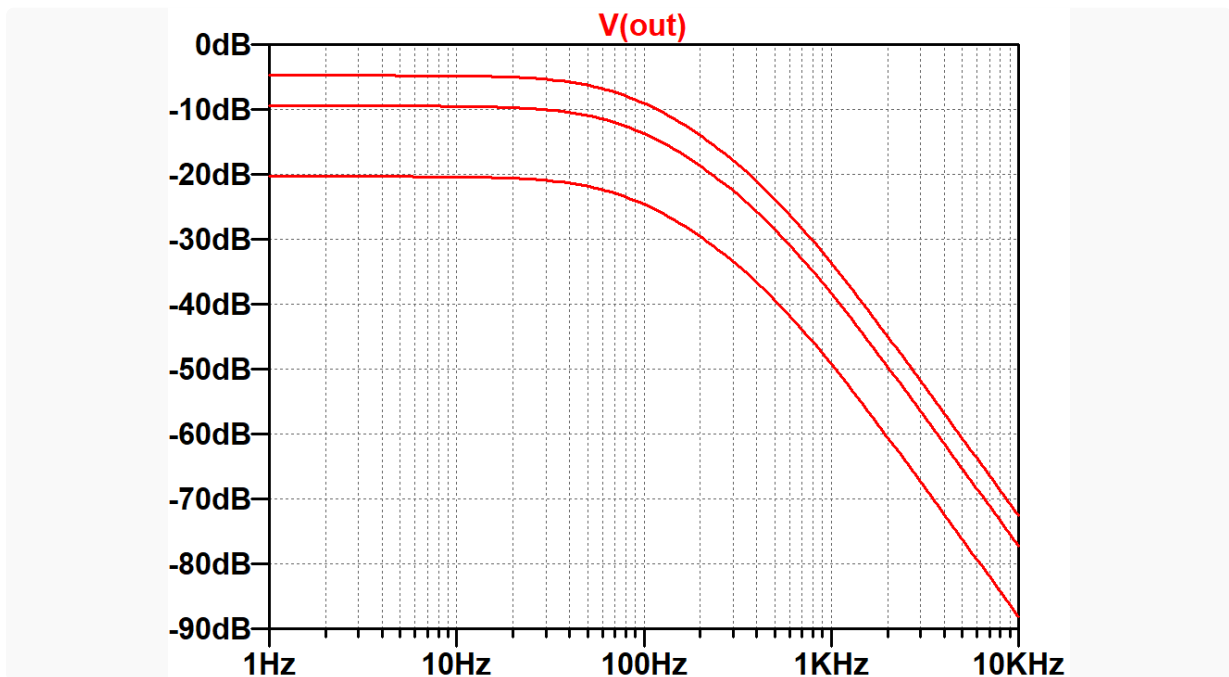
The bass control circuit consists of a cascade of two RC low-pass filters followed by a volume control.



The potentiometer and resistor below it effectively form a $600\text{k}\Omega$ control with a lower limit of $100\text{k}\Omega$. Here is the response based on a SPICE Simulation2 with the bass control at minimum, 50-percent resistance, and maximum.



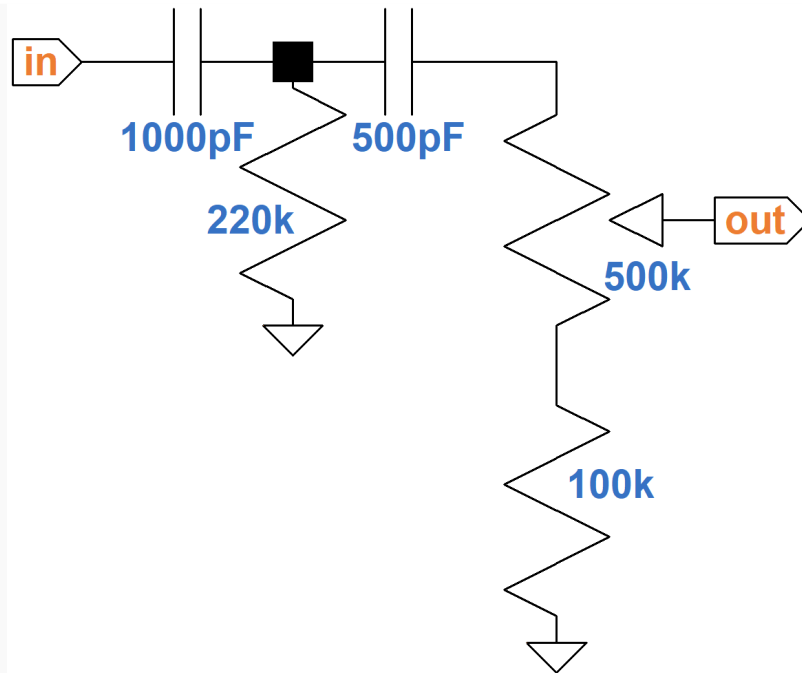
The driving circuit output impedance is assumed to be $139\text{k}\Omega$. Clearly there is severe treble cut, which ensures that the bass control only affects bass frequencies. Moreover, if we extend the plot down to frequencies well below audio, we see that parts values are chosen to put bass frequencies above the -5dB , -10dB , and -20dB low-frequency plateaus created when the capacitors act as open circuits.



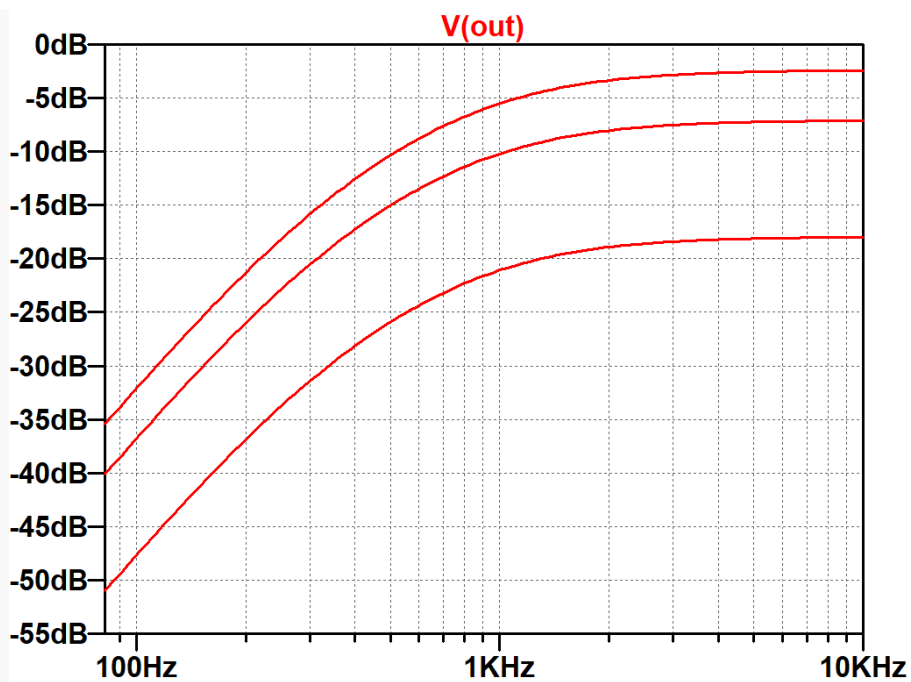
By the time the signal frequency increases to 82Hz, the lowest note on a guitar with standard tuning, response is about halfway to the -40dB per decade region created by the cascade of two [RC lowpass filters](#).

Treble Control

The treble control consists of a high-pass T network followed by a volume control.



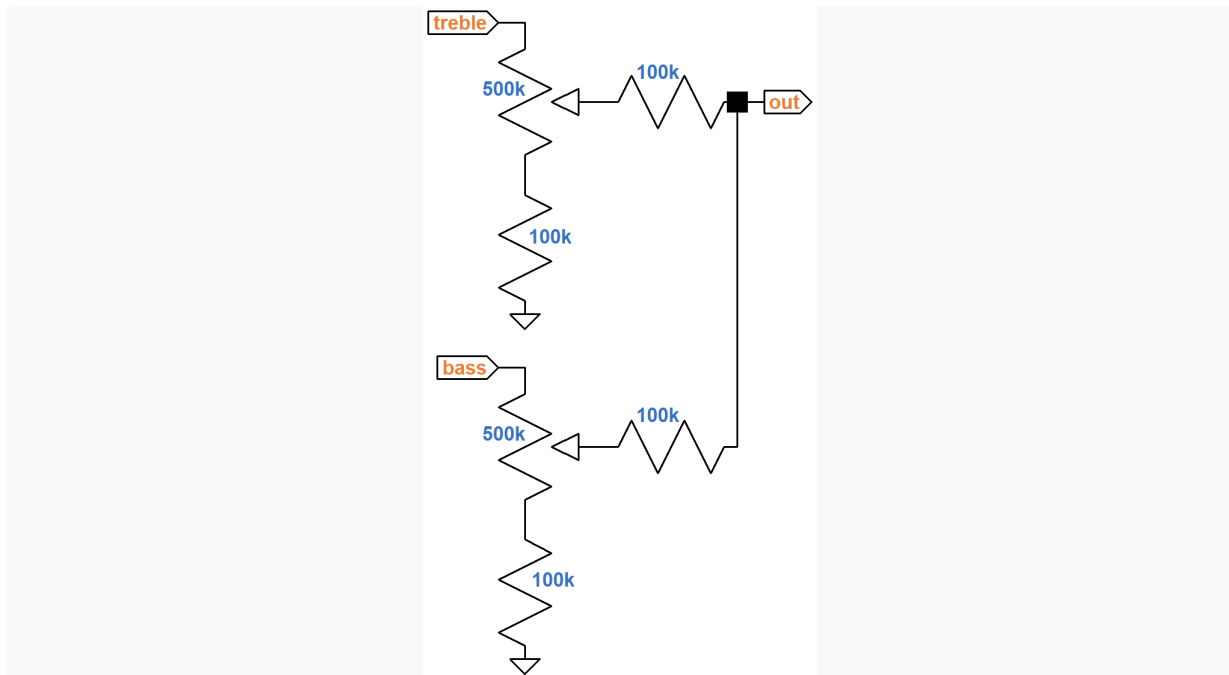
Here is the response assuming a driving circuit output impedance of 51kΩ.



This time the control region extends into the treble plateau.

The Passive Mixer

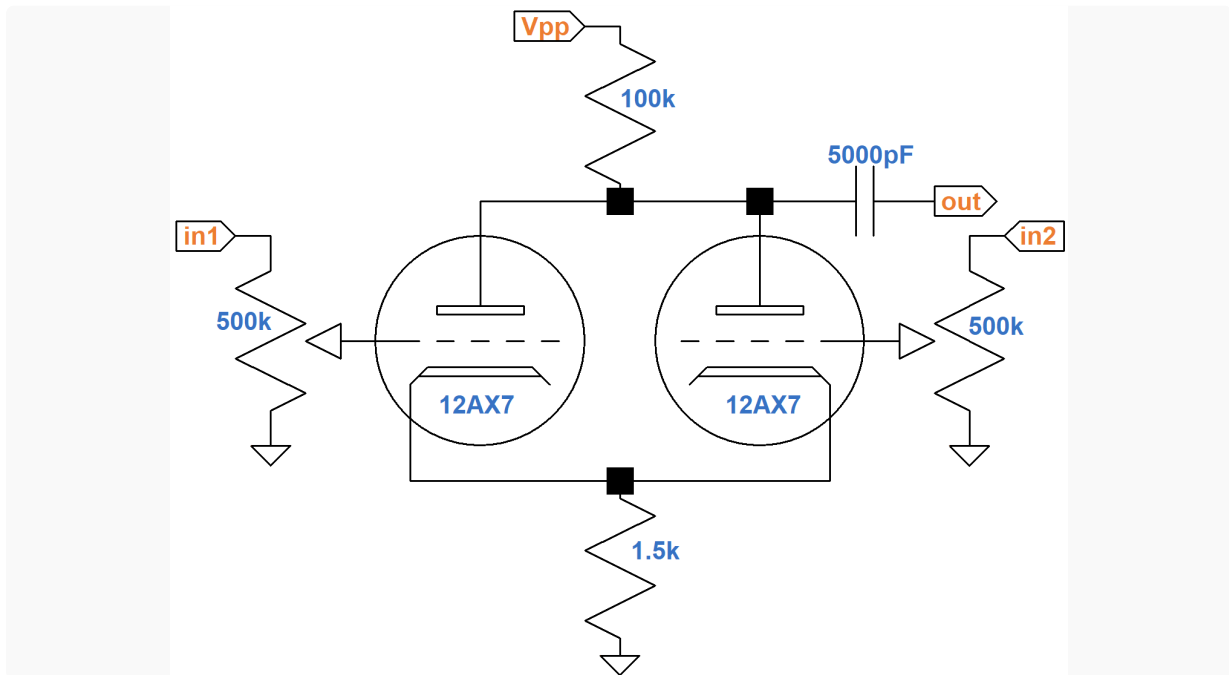
100k resistors are used to mix the treble and bass signals together.



In contrast to the isolation of the bass and treble driving stages, the passive mixer introduces some interactivity to the controls. Turning down the bass control affects treble response via the mixing resistors and vice versa. Even with isolated drivers, the Chet Atkins controls have substantial interactivity, a characteristic common to many classic amplifiers. The 100k Ω limiting resistors mitigate the effect somewhat by limiting the load each stage presents to the other.

Active Mixer

The Chet Atkins also contains an active mixer that is rarely seen in a guitar amp. The preamp has two channels, each with its own bass and treble controls and passive mixer. The channels are combined in a 12AX7-based mixing circuit.



The output is taken from where the plates connect to the 100k Ω plate load resistor. Since the resistors carry the plate current of two triodes, the DC operating point is the same as for a single triode with a 200k Ω plate load resistor and 3k Ω cathode resistor.

AC voltage gain for each channel is less than for a single triode amplifier. Imagine a rising voltage on the left input (channel 1) and no signal on the right. The increasing grid-to-cathode voltage causes an increase in plate current through the left triode, which creates more current through the cathode resistor. This raises the voltage across the resistor, causing the cathode voltage to rise and counteract the rising grid-to-cathode voltage induced by the input signal. This represents classic negative feedback. Moreover, there is common cathode coupling: a rise in cathode voltage for the left triode creates a similar rise in cathode voltage for the right triode. The net result is that the rising input signal causes the total current through the 100k Ω plate load resistor to increase while at the same time negative feedback in both triodes causes the total current to decrease.

The Radiotron Designer's Handbook³ calls this a "common plate load mixer." We posted a [calculator](#) that computes voltage gain and output impedance based on the tube and the values for the plate load and cathode resistors.

References

1Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 68-69.

2Richard Kuehnel, [Guitar Amplifier Electronics: Circuit Simulation](#), (Seattle: Amp Books, 2019).

3F. Langford-Smith, Ed., **Radiotron Designer's Handbook**, 4th Ed., (Harrison: RCA, 1953), p. 800.

3Richard Kuehnel, [Circuit Analysis of a Legendary Tube Amplifier: The Fender Bassman 5F6-A](#), 3rd ed., (Seattle: Amp Books, 2009).